

AtmoSound: An ML Framework for Venue-Adaptive Playlist Generation

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Abstract—Music is a measurable driver of customer perception, dwell times, and customer stickiness, yet most venues still curate playlists through intuition rather than data. AtmoSound addresses this gap by developing an end-to-end machine learning pipeline that ingests venue-level data from the Google Maps Places API and generates a Spotify-style playlist whose acoustic properties match the venue’s atmospheric identity. We implemented Multi-Output Ridge Regression, a two-layer Neural Network, and K-Means using NumPy, trained on 4,484 Manhattan venues with pseudo-labels derived from a genre-to-audio-profile mapping. Predictions are made in a 7-dimensional audio feature space (danceability, energy, acousticness, valence, instrumentality, liveness, speechiness) and songs are selected by matching the predicted profile against 112 genre centroids computed from 91,271 Spotify tracks. Ridge Regression, selected for deployment via a three-tier decision procedure, achieves a test-set MSE of 0.000281, cosine similarity of 0.9991, and Top-5 genre retrieval accuracy of 98.6%, substantially exceeding the mean-prediction baseline of MSE=0.004124 and Top-5=36.3%. The system is deployed as a Streamlit web application in which venue operators paste a Google Maps URL and receive a curated 20-track playlist together with interactive audio profile visualizations. We hope that our web application will enable venue owners to leave the guesswork out of curating a playlist for their business and instead use a machine-learning-enabled, data-driven approach to improve the customer experience and ultimately the customer’s desire to return. This work demonstrates that structured venue metadata can serve as a viable signal for music recommendation systems, reducing barriers to entry for data-driven curation tools across the hospitality industry.

I. INTRODUCTION

Background music directly affects how customers feel, how long they stay, and how much they spend [1], [2]. Research has shown

that when music is congruent with a venue’s atmosphere, customers report higher satisfaction and longer dwell times than when music is incompatible with the venue [2]. Despite this, most venues still select music manually or rely on genre-based B2B services such as Soundtrack, Your Brand, and Mood Media, which deliver standardized playlists curated by human editors without alignment to a specific venue’s identity. For example, a sports bar and an upscale Italian restaurant may be assigned the same genre channel even though they call for very different types of music.

Though prior work on music recommendation does not address this problem directly, collaborative filtering infers preferences from individual listening history, which venues do not have. Content-based approaches, such as the CNN model proposed by Van den Oord et al. [5], can recommend songs without listening history by training on raw audio features, and hybrid architectures [6] combine audio content with collaborative signals to further improve coverage. Context-aware systems have also begun incorporating location and situational factors [7]: Kaminskis et al. [8] matched music to points of interest using audio tags, and Wang et al. [9] jointly embeds audio content and contextual metadata. However, all of these systems are designed to optimize for individual listener preference. None of them are informed by the venue metadata that is already publicly available on platforms such as category, price level, other boolean attribute flags, and customer review text on Google Maps as a direct predictor of acoustic properties—this is the gap that this

project fills.

This paper presents *AtmoSound*, a machine learning pipeline that takes a Google Maps venue URL as input and generates a playlist matching the venue’s atmosphere. We implement two models from scratch in NumPy: **Multi-Output Ridge Regression**, chosen for its closed-form solution and interpretable weights in a correlated high-dimensional feature space, and a **two-layer Neural Network** ($128 \rightarrow 64 \rightarrow 7$, ReLU/Sigmoid), which captures non-linear feature interactions that Ridge cannot model. Performance is evaluated using **Mean Squared Error (MSE)**, **Cosine Similarity**, and **Top- K Genre Retrieval Accuracy**. The system is deployed as a Streamlit web application for venue owners and bar managers who want a data-driven playlist without music expertise: the user pastes a Google Maps URL and receives a 20-track playlist in under one second, with an audio profile visualization and busyness sliders for real-time adjustment.

AtmoSound shows that publicly available venue metadata can serve as a viable input signal for music recommendations, which is a direction that existing context-aware recommendation systems have not explored. For venues, it offers a simple and low-cost alternative to human-curated B2B music services. More broadly, this project democratizes access to sophisticated music curation, reducing dependence on expensive B2B services and enabling small venue operators to compete fairly. Beyond individual venues, this work demonstrates that public metadata can power recommendation systems across hospitality. Future validation across diverse geographies will strengthen generalizability.

II. BACKGROUND

A. Project Scope and Significance

This project addresses the problem of venue-adaptive music recommendations: given structured metadata describing a physical venue, generate a playlist whose acoustic properties match the venue’s atmosphere. Our motivation

comes from a well-established body of research on music and consumer behavior. Milliman [4] showed that slower-tempo background music in supermarkets increased both time in-store and total spend. A 2017 meta-analysis of 30 years of experiments confirmed that the effect holds across retail and dining contexts, and is strongest when music is congruent with other sensory attributes of a given location [1]. Wirtz and Mattila [2] showed that matched music-scent pairings produced better evaluations than either element alone. Taken together, these findings suggest that music selection is a meaningful and controllable driver of revenue and positive customer experience for venue operators.

B. Disputes and Common Knowledge

Despite this evidence, there is a gap between what research recommends and what the industry does. Academic work consistently finds that acoustic properties such as tempo, loudness, and valence affect customer behaviors [1], [4], but commercial music services continue to use broad genre-based channels curated by human editors, treating genre as an appropriate proxy for atmosphere. Douce et al. [3] found that venue managers who are aware of the music-atmosphere research still rely primarily on personal taste and intuition when making playlist decisions. This suggests the problem is not just a lack of tools but also a lack of operationalization. There is also an open question about the right level of acoustic granularity, as behavioral experiments typically study tempo and loudness [4], while Spotify’s API uses a broader composite 7-dimensional feature space (danceability, energy, acousticness, valence, instrumentalness, liveness, speechiness). AtmoSound uses the 7-dimensional space because it offers finer resolution and is available at scale from existing music data.

C. Comparisons to Prior Work

Music recommendation systems have developed in roughly three phases. Early collaborative filtering (CF) approaches infer user preferences from listening history and co-occurrence.

CF works well for personal recommendations on music streaming platforms but cannot be applied in the venue setting because no individual listening history exists. Van den Oord et al. [5] proposed training CNNs on raw audio spectrograms to produce cold-start recommendations without any listening history. Wang and Wang [6] extended this with hybrid models combining audio content and collaborative signals to improve coverage.

These content-based and hybrid approaches have advanced personalized recommendations, but are designed to predict what an individual listener wants to hear—not what acoustic signals a physical space should have. Murciego et al. [7] reviewed 128 papers on context-aware music recommendation and found that location and venue-level atmosphere are among the least studied contextual factors. Kaminskas et al. [8] proposed matching music to points of interest by auto-tagging audio tracks with semantic descriptors. Wang et al. [9] introduced CAME, which jointly embeds audio content and contextual metadata for music recommendations.

D. Knowledge Gaps and Positioning

None of the above approaches uses publicly available, structured venue metadata—such as primary type, price level, boolean amenity flags, and customer review text from Google Maps—as a direct input for predicting acoustic properties. Kaminskas et al. [8] is the closest mechanically, but it uses audio-side semantic tags rather than actual venue attributes and does not produce a continuous audio profile. CAME [9] incorporates contextual embeddings but requires user-item interaction data that venues do not have.

AtmoSound differs from these approaches in two ways: first, the input is the venue itself—structured Google Maps metadata is encoded into a feature vector and used to regress a 7-dimensional audio profile directly. Second, the architecture separates the problem into two stages: predict the audio profile, then retrieve songs from that point in acoustic space—making each stage independently interpretable. This

two-stage design also allows the retrieval step to be updated (e.g., via Spotify API) without retraining the regression model.

III. END-TO-END ML PIPELINE

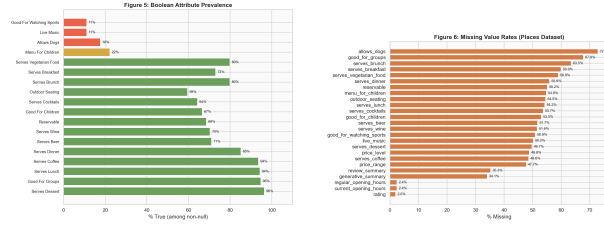
A. Data Collection and Exploration

Dataset 1: Manhattan Venues. We collected 4,484 venue records via the Google Maps Places API (New Places API v1) in April 2026. Each record contains 41 columns spanning four data types: (i) numerical—rating and geographic coordinates; (ii) categorical—primary type (175 unique values), price level (four ordinal tiers), and neighbourhood (33 Manhattan neighbourhoods); (iii) boolean—18 attribute flags including `live_music`, `outdoor_seating`, `allows_dogs`, and `good_for_watching_sports`; and (iv) free text—a generative summary produced by Google and a review summary aggregated from user reviews.

Exploratory Data Analysis. EDA is implemented in `eda_visualizations.py` and covers both datasets. Three findings directly shaped preprocessing and model design.

First, a bar chart of boolean attribute prevalence (Fig. 1a) reveals that the 18 flags fall into two groups: near-universal flags like `serves_dessert` (96% true) that carry no discriminative power, and rare atmospheric flags like `live_music` (11%) and `good_for_watching_sports` (11%) that are the strongest venue differentiators. A missing-value bar chart (Fig. 1b) shows these same high-signal flags are also the most frequently unreported (50–73% missing), which motivated the ternary encoding scheme (1/0/−1) rather than simple imputation.

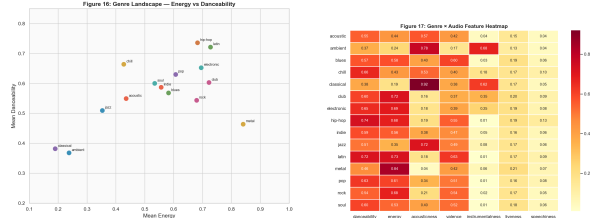
Second, a scatter plot of genre centroids in the energy–danceability plane (Fig. 2a) and a heatmap of genre-by-audio-feature means (Fig. 2b) confirm that genres occupy distinct, spatially separated regions of the 7-dimensional audio feature space. This separation validates the core retrieval architecture: nearest-centroid lookup produces qualitatively different results



(a) Boolean Attribute Prevalence (b) Missing Value Rates

Fig. 1: EDA findings for the Manhattan Venues dataset. High-signal boolean flags (left) are also the most frequently unreported (right).

for different genre targets, making the approach reliable.



(a) Genre Landscape: Energy vs. Danceability (b) Genre × Audio Feature Heatmap

Fig. 2: Genre centroids occupy distinct regions in the 7-dimensional audio feature space, validating the nearest-centroid retrieval strategy.

Third, a correlation matrix of the Spotify audio features (Fig. 3) shows that energy and loudness correlate strongly ($r = 0.76$) while acousticness and energy are inversely related ($r = -0.73$). Popularity is uncorrelated with all audio features, which supports using it as a within-genre sampling weight rather than as a model input.

Dataset 2: Spotify Tracks. We used the Kaggle Spotify Tracks Dataset containing 114,000 tracks spanning 114 genres (1,000 per genre), each annotated with seven audio features computed by Spotify’s API. After filtering tracks with popularity below 10, 91,271 tracks remain. Near-duplicate genres (singer-songwriter / songwriter, Euclidean distance 0.000; latino / reggaeton, distance 0.019) are merged, yielding

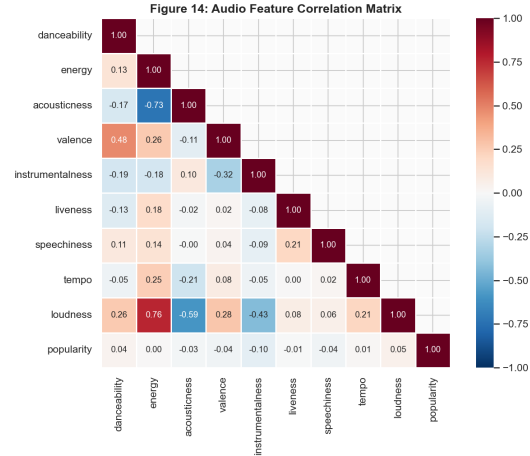


Fig. 3: Audio Feature Correlation Matrix. Energy and loudness are strongly correlated ($r = 0.76$); acousticness and energy are inversely related ($r = -0.73$).

a finalized 112 genre centroids used for all retrieval operations.

B. Preprocessing Pipeline

All preprocessing is implemented in `AtmoSound_Preprocessing.ipynb`.

Boolean encoding treats missingness as a third informative state: each flag is encoded as 1 (true), 0 (false), or -1 (unknown).

Rating and price level: the 2% of venues with missing ratings are imputed with the training median (4.5); price level is ordinally encoded 1–4 with a separate binary missing indicator column.

Categorical features: neighbourhood (33 values) and primary type (175 values) are one-hot encoded, producing 208 additional binary features; unseen values at inference time default to the zero vector.

Text features: review summary and generative summary are concatenated per venue, then TF-IDF vectorized (unigrams and bigrams, sublinear TF, L2 normalization) and reduced to 50 dense dimensions via Truncated SVD, capturing thematic signals such as “lively,” “cozy,” or “upscale.” The final feature matrix is $4,484 \times 279$ and all features are min-max normalized to $[0, 1]$.

Pseudo-label construction. Since no ground-truth venue-playlist pairs exist, pseudo-labels are constructed in two steps. First, each venue’s primary type is mapped to a genre group (e.g., `italian_restaurant` $\rightarrow$ `jazz/classical/soul`; `bar` $\rightarrow$ `jazz/blues/rock`), and the mean Spotify audio profile of that group becomes the base target. Second, boolean flags and price level modulate this base: `live_music=True` blends 15% toward `jazz/blues/rock/folk`; `outdoor_seating=True` blends 8% toward `folk/acoustic/indie`; very expensive venues receive -0.04 on energy and $+0.06$ on acousticness. All values are clipped to $[0, 1]$, producing a $4,484 \times 7$ label matrix.

C. Model Implementation

All models are implemented from scratch in NumPy without any ML libraries. The task is formulated as **multi-output regression**: given a venue feature vector $\mathbf{x} \in \mathbb{R}^{279}$, predict a target audio profile $\mathbf{y} \in [0, 1]^7$, where each of the 7 dimensions corresponds to one Spotify audio feature (danceability, energy, acousticness, valence, instrumentality, liveness, speechiness). The choice of regression over classification is motivated by the continuous nature of audio features: small differences in predicted energy or acousticness translate to meaningfully different genre matches in the retrieval step.

Inputs and Outputs. All three models share the same 279-dimensional min-max normalized input vector encoding rating, price level, 18 boolean attributes, 33 neighbourhood indicators, 175 primary type indicators, and 50 TF-IDF/SVD text dimensions. The output for Ridge and the Neural Network is a 7-dimensional vector $\mathbf{y} \in [0, 1]^7$. K-Means outputs a cluster assignment, from which the centroid’s stored audio profile is used as the prediction.

Method 1: Multi-Output Ridge Regression. The problem is framed as minimizing the regularized Frobenius-norm objective over all 7 targets simultaneously:

$$\mathcal{L}(W) = \|XW - Y\|_F^2 + \lambda \|W\|_F^2 \quad (1)$$

where $X \in \mathbb{R}^{n \times (d+1)}$ is the feature matrix augmented with a bias column, $Y \in \mathbb{R}^{n \times 7}$ is the pseudo-label matrix, and $W \in \mathbb{R}^{(d+1) \times 7}$ contains the weights for all 7 output dimensions. The closed-form solution is:

$$W^* = (X^\top X + \lambda I)^{-1} X^\top Y \quad (2)$$

The bias row is excluded from regularization by zeroing out the corresponding diagonal entry of λI before solving. Inference is $\hat{\mathbf{y}} = [x, 1] W^*$, a single matrix multiplication. Ridge is well-suited to this problem because the feature space contains many correlated one-hot columns; L2 regularization shrinks them jointly rather than eliminating them arbitrarily, and the resulting weight matrix $W_{:,j}^*$ is directly interpretable—each column shows which features drive a given audio dimension.

Method 2: Two-Layer Neural Network. The network learns a non-linear mapping from venue features to audio profile. The forward pass is:

$$z^{(1)} = XW^{(1)} + b^{(1)}, \quad a^{(1)} = \max(0, z^{(1)}) \quad (3)$$

$$z^{(2)} = a^{(1)}W^{(2)} + b^{(2)}, \quad a^{(2)} = \max(0, z^{(2)}) \quad (4)$$

$$\hat{\mathbf{y}} = \sigma(a^{(2)}W^{(3)} + b^{(3)}), \quad \sigma(z) = \frac{1}{1 + e^{-z}} \quad (5)$$

with layer dimensions $W^{(1)} \in \mathbb{R}^{279 \times 128}$, $W^{(2)} \in \mathbb{R}^{128 \times 64}$, $W^{(3)} \in \mathbb{R}^{64 \times 7}$. The training objective is MSE over all 7 outputs:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \|\hat{y}_i - y_i\|^2 \quad (6)$$

Gradients are computed via backpropagation and weights are updated with the Adam optimizer. The sigmoid output layer naturally bounds all predictions to $[0, 1]$, matching the normalized target space without post-hoc clipping. The neural network captures non-linear interactions between features that Ridge cannot—for example, the combined effect of `live_music=True` and

price_level=expensive, which should shift the profile toward jazz rather than rock, requires a conditional relationship that a linear model cannot represent.

Method 3: K-Means Clustering (Interpretive). K-Means partitions venues into K clusters by minimizing the within-cluster sum of squared distances:

$$\mathcal{L} = \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2 \quad (7)$$

where μ_k is the centroid of cluster C_k . Each cluster is assigned the mean audio profile of its members as its prediction. K-Means reveals natural venue groupings without depending on pseudo-label quality, providing an interpretable set of venue archetypes (e.g., upscale cocktail bar, casual brunch spot) that can be inspected independently of the regression models. Cluster count $K = 25$ is selected by silhouette score.

D. Playlist Retrieval

Given predicted profile $\hat{y}$, the top-5 nearest genre centroids are selected by Euclidean distance. Songs are sampled from each genre proportionally to inverse distance weights, with Spotify popularity as the within-genre sampling probability. The resulting 20-track playlist is sorted into an energy arc (building to a midpoint peak, then cooling), producing a natural atmospheric progression appropriate for a continuous venue service period.

IV. MODEL EVALUATION

A. Training and Validation

The 4,484-venue dataset is partitioned using a two-stage stratified split on primary type (seed=42), yielding an overall 64/16/20 train/validate/test split. In the first stage, 80% of the full dataset (3,587 samples) is held out as the train-plus-validation pool, and the remaining 20% (897 samples) becomes the held-out test set. In the second stage, the 3,587-sample pool is split 80/20 again, producing a train subset of 2,869 samples (64%) and a validation subset of 718 samples (16%). Stratification

is applied at both stages to ensure that rare venue categories appear in all three splits in proportional representation. All pseudo-labels are constructed before any splitting to prevent leakage. All preprocessing transformers (TF-IDF, SVD, MinMax scaler) are fitted on the 2,869-sample train subset only and applied without refitting to the validation and test sets.

B. Hyperparameter Tuning

Ridge Regression: A grid search is run over $\lambda \in \{0.001, 0.01, 0.1, 1, 10, 100\}$. The optimal value is $\lambda = 0.1$, which balances fit quality against coefficient shrinkage. Values below 0.01 show marginal validation improvement but increase sensitivity to correlated one-hot columns; values above 1 over-regularize and degrade Top-5 accuracy by 2–3 percentage points.

Neural Network: A grid search is run over learning rate $\alpha \in \{0.001, 0.005, 0.01\}$, batch size $\in \{32, 64, 128\}$, and dropout rate $\in \{0.0, 0.2\}$ (18 configurations total). Each configuration is trained for up to 500 epochs and evaluated on validation MSE. The best configuration uses $\alpha = 0.01$, batch size 32, and dropout 0.0.

K-Means: The number of clusters K is selected by silhouette score over $K \in \{10, 15, 20, 25, 30\}$. $K = 25$ achieves the highest silhouette score and produces the most semantically distinct venue archetypes.

C. Overfitting and Underfitting

Ridge Regression is inherently protected against overfitting by its L2 penalty, which shrinks correlated coefficients jointly. Underfitting is confirmed absent by comparing training and validation MSE: at $\lambda = 0.1$, both values are within 0.00005 of each other.

For the Neural Network, overfitting is addressed via dropout (rate 0.2 after each hidden layer) and early stopping (patience=20), which halts training when validation MSE stops improving and restores weights from the best epoch. No divergence between training and validation loss curves was observed in the selected configuration.

D. Metrics

No single metric captures all the ways a predicted audio profile can fail, so we use three metrics that each address a distinct failure mode.

Mean Squared Error (MSE) measures absolute prediction error across all seven dimensions. Since all targets are in $[0, 1]$, MSE is directly comparable across dimensions and models. Its limitation is that low-variance dimensions like liveness ($\sigma = 0.045$) and speechiness ($\sigma = 0.030$) contribute little even when the prediction direction is correct.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i^{\text{true}} - y_i^{\text{pred}})^2 \quad (8)$$

where, n = samples; y_i^{true} = true audio profile (7D: acousticness, energy, instrumentalness, liveness, speechiness, tempo, valence); y_i^{pred} = predicted profile.

Cosine Similarity (CosSim) evaluates profile *shape* rather than absolute values. This matters because genre retrieval is direction-sensitive: the nearest centroid is determined by whether the model correctly identifies a high-energy/low-acousticness profile versus the reverse.

$$\text{CosSim} = \frac{1}{n} \sum_{i=1}^n \frac{y_i^{\text{true}} \cdot y_i^{\text{pred}}}{\|y_i^{\text{true}}\| \|y_i^{\text{pred}}\|} \quad (9)$$

where, $\cdot$ = dot product; $\|\cdot\|$ = Euclidean norm; n = samples.

Accuracy@K (Acc@K) is the most direct measure of end-user value. A model can achieve low MSE and high Cosine Similarity yet still place the predicted profile near the wrong genre centroid when boundaries are close.

$$\text{Acc@K} = \frac{1}{n} \sum_{i=1}^n \mathbf{1} \left[\arg \min_j \|c_j - y_i^{\text{true}}\| \in \text{TopK}(y_i^{\text{pred}}) \right] \quad (10)$$

where, c_j = genre centroid j ; $\text{TopK}(y_i^{\text{pred}}) = K$ nearest centroids to prediction; $\mathbf{1}[\cdot]$ = indicator (1 if true, 0 otherwise); $\arg \min_j$ = nearest centroid index to true profile.

Together they form a hierarchy: MSE checks regression accuracy, Cosine Similarity checks directional correctness, and Top- K checks real-world retrieval utility.

E. Results

TABLE I: Test-Set Performance Across Models

Model	MSE	CosSim	Top-3	Top-5
Mean Baseline	0.004124	0.9874	29.2%	36.3%
K-Means ($K=25$)	0.001184	0.9964	85.3%	88.3%
Neural Network	0.000255	0.9992	97.8%	98.3%
Ridge ($\lambda=0.1$)	0.000281	0.9991	98.0%	98.6%

Baseline comparison. The mean-prediction baseline achieves MSE 0.004124 and Top-5 accuracy of only 36.3%, representing a model with no venue-specific signal. Ridge reduces MSE by 93% and improves Top-5 from 36.3% to 98.6%, confirming that venue metadata carries genuine predictive signal.

K-Means vs. regression methods. K-Means improves meaningfully over the baseline (Top-5 88.3%) but lags behind both regression models by approximately 10 percentage points. This gap is attributable to prediction granularity: every venue within a cluster receives an identical audio profile, so K-Means cannot distinguish venues of the same type that differ in price level, boolean attributes, or review text.

Ridge vs. Neural Network. The two regression models perform at near-identical levels. The Neural Network achieves slightly lower MSE (0.000255 vs. 0.000281) but Ridge leads on Top-5 accuracy (98.6% vs. 98.3%). All differences are within 0.3 percentage points. The expected advantage of the Neural Network—modeling non-linear feature interactions—does not materialize, likely because pseudo-label circularity makes the target function close to linear in the one-hot encoded `primary_type` input, leaving little room for non-linear modeling to add value.

Per-metric interpretation. High Cosine Similarity scores across all learned models (0.9964–0.9992) indicate that even K-Means produces profiles pointing in the correct direction in audio space. The sharper discrimination appears in Top- K accuracy, where the gap between K-Means (88.3%) and the regression models (98.3–98.6%) confirms that directional correctness

alone is insufficient—precise centroid placement also matters for retrieval quality.

Ridge is selected via a three-tier procedure. The Top-5 gap (0.3 pp) and cosine gap (0.0001) both fall within the 3% margin defined as inconclusive, so the practical tiebreaker applies: Ridge’s single matrix-multiplication inference satisfies Streamlit’s low-latency requirements, and its weight matrix is directly interpretable per output dimension. A feature ablation shows that categorical features alone achieve 98.6% Top-5 accuracy, nearly matching the full 279-feature vector. Boolean flags alone and TF-IDF text alone each reach approximately 70–74%, confirming that `primary_type` is the dominant predictor.

V. MODEL DEPLOYMENT AND FRONT END APPLICATION

A. Application Overview

AtmoSound is deployed as a Streamlit web application that allows venue operators to generate a venue-matched playlist by pasting a Google Maps URL. The app is organized into six pages—Home, Statistics, Playlist, About, Favorites, and Recent—and is designed for non-technical users such as restaurant managers, hotel operators, and bar owners who want data-driven music recommendations without requiring any domain expertise.

B. Model Selection

Ridge Regression is selected as the deployed model based on the three-tier decision procedure described in Section IV-E. Because neither Top-5 accuracy nor Cosine Similarity decisively favor one model, the tiebreaker is practical suitability: Ridge inference is a single matrix multiplication, making it substantially faster than the Neural Network’s forward pass, and its weight matrix is directly interpretable. The Neural Network requires more computation and offers no interpretability advantage, so Ridge is the clear choice for deployment.

C. Target Population

The application is designed for venue owners, bar managers, and hotel operators who need background music that fits their space but lack the time or expertise to curate it manually. Current alternatives—manual playlist curation or subscription-based B2B services—either require music knowledge or deliver standardized content that is not tailored to a specific venue. AtmoSound gives this group a free, instant alternative: they provide a Google Maps link that already describes their venue, and the system handles the rest. The busyness sliders (7 am to 2 am) add a practical layer of control, letting operators adjust the playlist in real time to match the current energy in the room.

D. User Interface and Outputs

The **Home** page (Fig. 4) contains a single text input field for a Google Maps URL and a Generate Playlist button. Clicking the button triggers the full pipeline: the app resolves the Place ID, fetches venue data from the Google Maps Places API, builds the 279-dimensional feature vector, runs Ridge Regression to predict the audio profile, matches it to the nearest genre centroids, and samples 20 tracks. All of this happens in the back end without further user input.

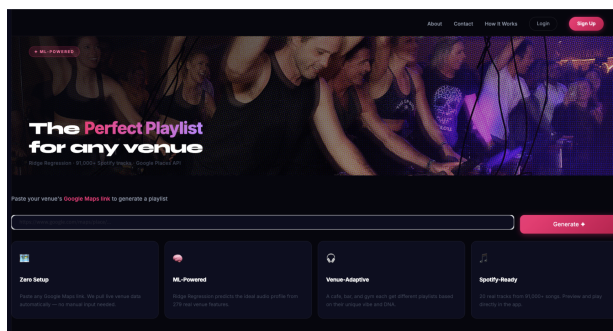


Fig. 4: AtmoSound Home page. Venue operators paste a Google Maps URL and click Generate to receive a 20-track playlist in under one second.

The **Statistics** page (Fig. 5) displays results in two columns. The left column shows three metric cards (rating, review count, and Energy

Score), Vibe Tags derived from the venue’s active attributes and top genre, a Sentiment Breakdown panel with color-coded progress bars for valence, energy, danceability, and acousticness, and a list of the top predicted genres. The right column shows the Acoustic Targets panel with the predicted value for each of the seven audio dimensions, and six Busyness by Hour sliders (7 am–2 am). When the operator adjusts the sliders and clicks Apply, the audio profile is shifted accordingly and the playlist is regenerated without re-running the model.

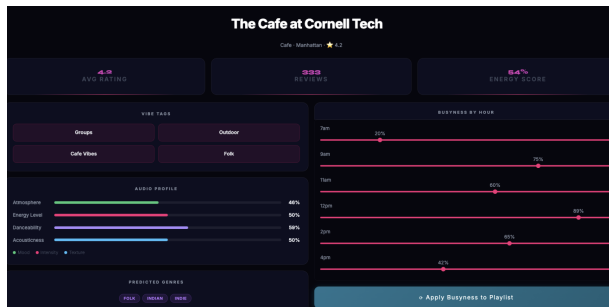


Fig. 5: Statistics page showing venue metrics, Vibe Tags, Sentiment Breakdown, predicted audio profile, and Busyness by Hour sliders.

The **Playlist** page (Fig. 6) lists the 20 recommended tracks. Each row shows the track name, artist, genre tag, and key audio features. For tracks with a valid Spotify ID, an embedded player is injected inline so users can preview songs directly within the app. Users can save the playlist to Favorites, which persists across sessions in a local JSON file.



Fig. 6: Playlist page showing the 20 recommended tracks with inline Spotify preview players.

E. Front-End and Back-End Connection

The front and back end are connected through a single cached pipeline object loaded at startup via `@st.cache_resource`. When the user clicks Generate on the Home page, `run_pipeline()` calls `AtmoSoundPipeline.generate_playlist()`, which runs the full chain—feature engineering, Ridge prediction, genre matching, and song sampling—and returns a structured result stored in `st.session_state`. All subsequent pages read from this session state; no additional model calls are made when navigating between pages. The Busyness adjustment is the one exception: clicking Apply re-runs only the genre matching and song sampling steps with the adjusted audio profile, not the Ridge model itself. This keeps the app responsive while still allowing real-time interaction.

VI. CONCLUSION

A. Summary of Goals, Approach, and Results

This project set out to answer whether publicly available venue metadata from Google Maps can serve as a reliable input for generating acoustically appropriate playlists. We built an end-to-end ML pipeline that encodes venue features into a 279-dimensional feature vector and uses Multi-Output Ridge Regression to predict a 7-dimensional audio profile. A Neural Network was implemented as an alternative, and K-Means clustering was used as an interpretive baseline. Songs are retrieved by matching the predicted profile to 112 Spotify genre centroids and sampling 20 tracks weighted by genre proximity and popularity.

Ridge Regression achieves a test-set MSE of 0.000281 and Top-5 Genre Retrieval Accuracy of 98.6%, substantially outperforming the mean-prediction baseline (MSE 0.004124, Top-5 36.3%). The Neural Network performs comparably but offers no decisive advantage, thus Ridge is selected for deployment. Feature ablation shows that `primary_type` is the dominant predictor, with boolean flags and text

features contributing only marginally on their own.

The most important caveat is *pseudo-label circularity*: because pseudo-labels are constructed by mapping `primary_type` to genre groups, and the model uses `primary_type` as an input feature, the model is partly recovering its own labeling rules rather than learning a true venue-to-atmosphere relationship. The reported metrics are therefore optimistic relative to real-world performance. Boolean attribute coverage also limits differentiation in practice—50–73% of attributes are unreported for any given venue, so two venues of the same category but different characters may receive similar playlists. Finally, the song pool is a fixed snapshot of 91,271 tracks, and the model is trained only on Manhattan venues, which constrains geographic generalizability.

B. Implications for Users and Society

For the target users, AtmoSound offers a practical alternative to manual curation or expensive B2B subscription services. Instead of relying on intuition or standardized genre channels, operators can input their venue’s existing Google Maps listing and receive an acoustically tailored playlist in under one second. More broadly, the project demonstrates that the structured metadata that venues already publish publicly is a usable signal for acoustic recommendation—a finding that could reduce the cost of entry for music curation tools across the hospitality and retail industries.

C. Future Work

The most pressing next step is addressing pseudo-label circularity by collecting human-annotated ground truth: surveying venue managers about their preferred playlists or running user studies would allow the model to be evaluated and retrained on real labels rather than rule-derived proxies. On the technical side, connecting to the live Spotify Web API would keep the song pool current. Extending the training data beyond Manhattan would improve geographic generalizability and reduce the model’s

reliance on Manhattan-specific neighbourhood encodings. Finally, incorporating time-of-day as a structured input feature, rather than relying solely on the busyness sliders, could allow the model to learn day-part acoustic preferences directly from data.

VII. CODE, DOCUMENTATION, AND DEPLOYMENT

- **GitHub:** <https://github.com/ML-Team-2026/AtmoSound-Final>
- **Streamlit App:** <https://atmosound-eejsk4pwner3sfjdojyujg.streamlit.app/>

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APPENDIX

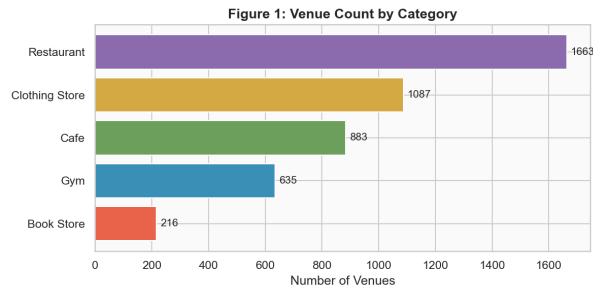


Fig. 7: Venue category counts in the Manhattan dataset.

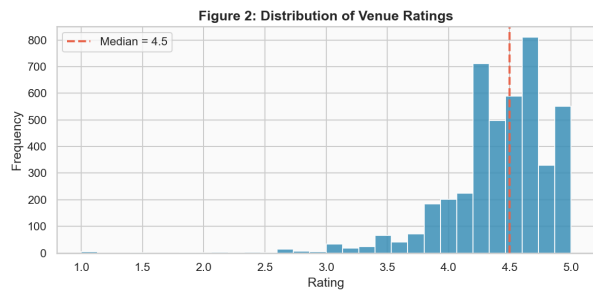


Fig. 8: Distribution of venue ratings.

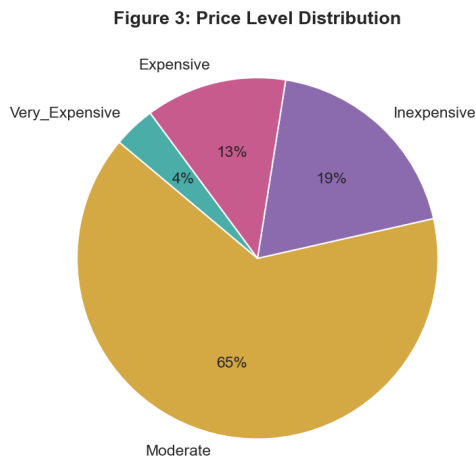


Fig. 9: Price level distribution across venues.

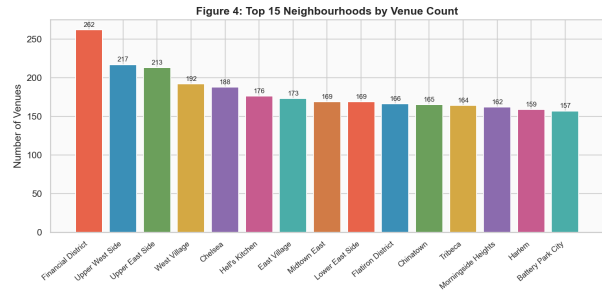


Fig. 10: Venue counts by Manhattan neighbourhood.

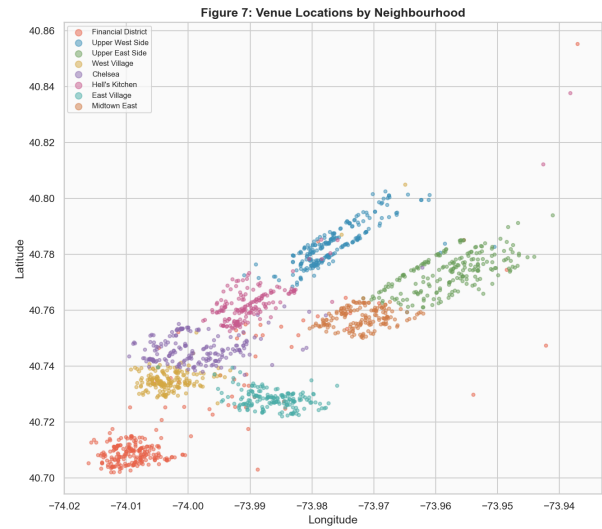


Fig. 11: Geographic scatter of venues across Manhattan.

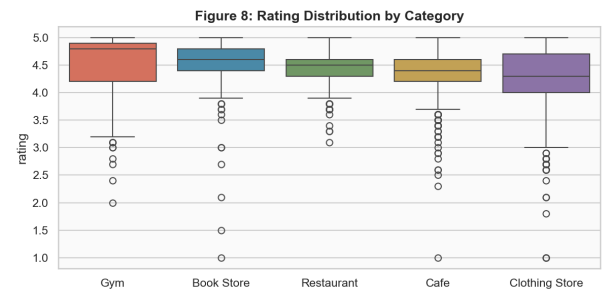


Fig. 12: Rating distribution by venue category (boxplot).

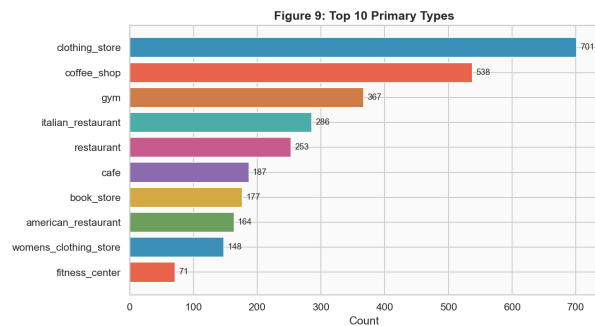


Fig. 13: Top primary venue types in the dataset.

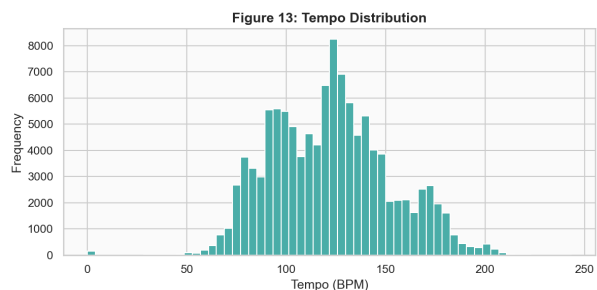


Fig. 17: Tempo distribution in the Spotify dataset.

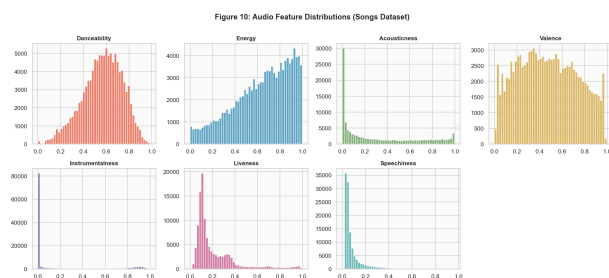


Fig. 14: Histograms of the seven Spotify audio features.

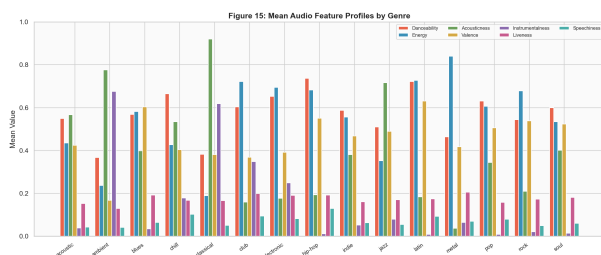


Fig. 18: Mean audio profiles for each of the 112 genre centroids.

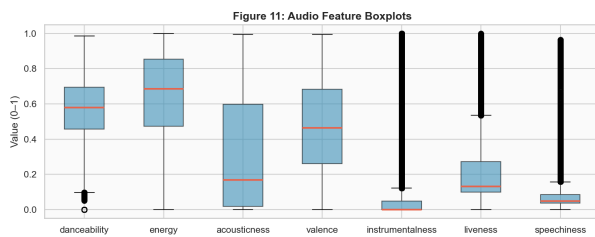


Fig. 15: Boxplots of Spotify audio features by genre group.

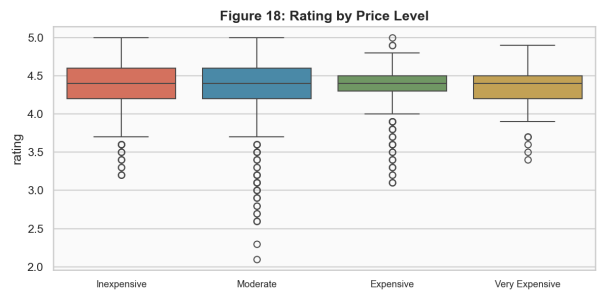


Fig. 19: Venue rating distribution broken down by price level.

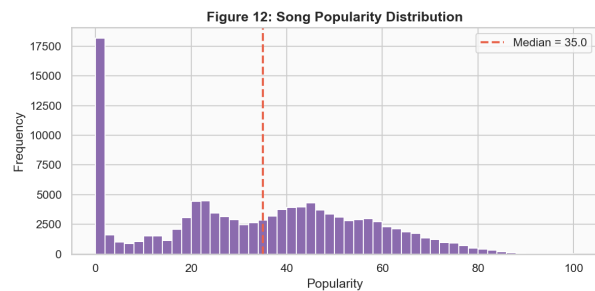


Fig. 16: Track popularity distribution in the Spotify dataset.

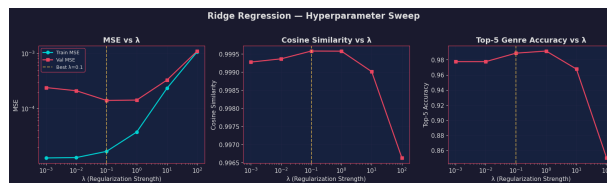


Fig. 20: Ridge Regression hyperparameter sweep: validation MSE vs. λ .

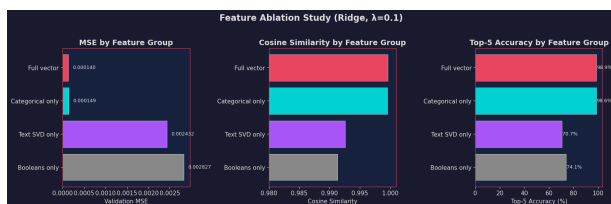


Fig. 21: Feature ablation study showing Top-5 accuracy for each feature group.



Fig. 25: Final model comparison on the held-out test set.

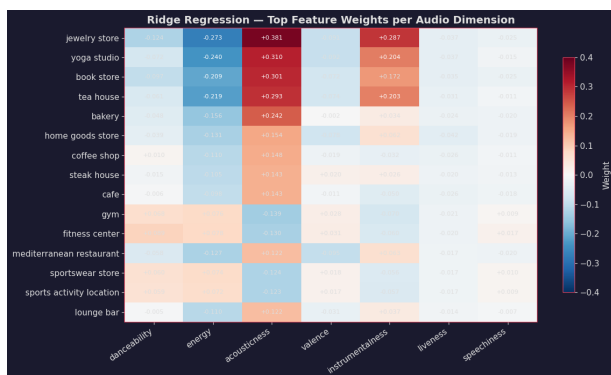


Fig. 22: Ridge Regression: top feature weights per audio dimension.

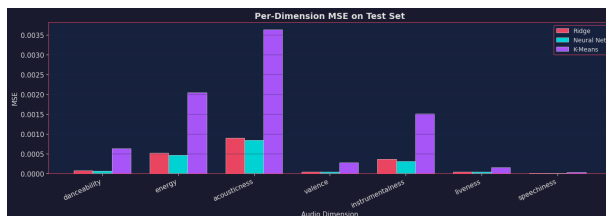


Fig. 26: Per-dimension MSE on the test set for Ridge, Neural Network, and K-Means.

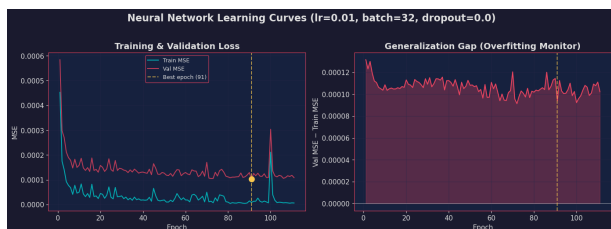


Fig. 23: Neural Network training and validation loss curves.



Fig. 24: K-Means cluster selection: silhouette score vs. K .

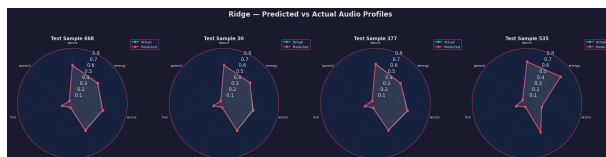


Fig. 27: Ridge Regression: predicted vs. actual audio profiles on test venues.